AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Meta Henrico Data Center, VA -- station KRIC, America/New_York
Building	OSM 1132541700 Meta Henrico Data Center
Facade gap	not applicable -- one building, no second facade
Weather record	43,736 real hourly records from KRIC
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-03-30 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. The reactive incumbent took 11 hours -- 10 more -- but declared 1 unsafe hour against the agent's 1. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	1 of 24 hours free cooling, 1 mode change(s), 1 unsafe hour(s)
Incumbent	11 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	6.670	18.0	6.670	0.743

01	mechanical	yes	switch budget	5.835	18.0	5.000	0.650
02	mechanical	yes	switch budget	6.390	18.0	5.560	0.852
03	mechanical	yes	switch budget	5.830	18.0	4.440	0.801
04	mechanical	yes	switch budget	5.000	18.0	4.440	0.756
05	mechanical	yes	switch budget	4.725	18.0	3.330	0.701
06	mechanical	yes	switch budget	4.720	18.0	3.890	0.776
07	mechanical	yes	switch budget	7.220	18.0	7.220	0.944
08	mechanical	yes	switch budget	11.110	18.0	12.780	1.561
09	mechanical	yes	switch budget	13.895	18.0	15.560	1.824
10	mechanical	yes	switch budget	17.500	18.0	18.890	1.893
11	mechanical	no	dry-bulb	20.560	18.0	21.110	1.439
12	mechanical	no	dry-bulb	22.225	18.0	22.780	1.347
13	mechanical	no	dry-bulb	23.615	18.0	23.890	0.960
14	mechanical	no	dry-bulb	24.445	18.0	24.440	0.838
15	mechanical	no	dry-bulb	25.560	18.0	25.560	1.006
16	mechanical	no	dry-bulb	25.280	18.0	25.000	1.040
17	mechanical	no	dry-bulb	24.720	18.0	24.440	1.127
18	mechanical	no	dry-bulb	24.445	18.0	23.330	1.416
19	mechanical	no	dry-bulb	22.500	18.0	20.560	1.476
20	mechanical	no	dry-bulb	20.555	18.0	17.220	1.528
21	mechanical	no	dry-bulb	20.835	18.0	18.890	1.252
22	mechanical	no	dry-bulb	20.000	18.0	19.440	1.169
23	FREE	yes	-	17.775	18.0	18.330	0.851

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.670 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.835 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.830 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.725 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.720 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.220 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.895 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.560 C against a 18.0 C limit, so it fails by 2.560 C. It would take a limit 2.560 C higher, or a bound 2.560 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.225 C against a 18.0 C limit, so it fails by 4.225 C. It would take a limit 4.225 C higher, or a bound 4.225 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.615 C against a 18.0 C limit, so it fails by 5.615 C. It would take a limit 5.615 C higher, or a bound 5.615 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.445 C against a 18.0 C limit, so it fails by 6.445 C. It would take a limit 6.445 C higher, or a bound 6.445 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.560 C against a 18.0 C limit, so it fails by 7.560 C. It would take a limit 7.560 C higher, or a bound 7.560 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.280 C against a 18.0 C limit, so it fails by 7.280 C. It would take a limit 7.280 C higher, or a bound 7.280 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.720 C against a 18.0 C limit, so it fails by 6.720 C. It would take a limit 6.720 C higher, or a bound 6.720 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.445 C against a 18.0 C limit, so it fails by 6.445 C. It would take a limit 6.445 C higher, or a bound 6.445 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.500 C against a 18.0 C limit, so it fails by 4.500 C. It would take a limit 4.500 C higher, or a bound 4.500 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.555 C against a 18.0 C limit, so it fails by 2.555 C. It would take a limit 2.555 C higher, or a bound 2.555 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.835 C against a 18.0 C limit, so it fails by 2.835 C. It would take a limit 2.835 C higher, or a bound 2.835 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.000 C against a 18.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.775 C, which is 0.225 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.851 C, and the margin is measured, not chosen: 0.851 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by. *** THIS WAS WRONG: the intake actually reached 18.330 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

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